THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** one or more new substances	Chemical changes form new substances.
2	**(d)** sour	Acids taste sour.
3	**(d)** malleability	Malleability.
4	**(a)** 6	Carbon has 6 protons.
5	**(d)** autotrophic	Plants make their own food.
6	**(d)** neuron	Neuron.
7	**(c)** produce new individuals like themselves	Basic definition.
8	**(a)** heredity and variation	Genetics deals with inheritance.
9	**(c)** straight lines	Rectilinear propagation.
10	**(a)** optical instrument	It forms images like a camera.
11	**(c)** charge	I = Q/t.
12	**(d)** magnetic field	Definition.
13	**(a)** biotic components	Living things.
14	**(a)** combination reaction	Two reactants form one product.
15	**(b)** 7	Neutral solutions have pH 7.
16	**(b)** gold	Gold is at the bottom of the series.
17	**(c)** ethane	Alkane.
18	**(c)** bile	Bile juice.
19	**(c)** insulin	Insulin.
20	**(d)** pollen grains	Anther is the pollen-making part.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Mass can neither be created nor destroyed in a chemical reaction (1); Total mass of reactants equals total mass of products (1).	1+1
22	Substance giving H^+ ions in water (1); HCl, H_2SO_4 (1).	1+1
23	Malleability: beaten into sheets (1); Ductility: drawn into wires (1).	1+1
24	Valency is 4 (1); It has 4 electrons in its outer shell (1).	1+1
25	Process of obtaining and using food (1); Autotrophic and heterotrophic (1).	1+1
26	Nerve cell (1); Dendrite, cell body, axon (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Count atoms: 1 Na, 2 Cl on the left, 1 Na, 1 Cl on the right (1); Put 2 before NaCl and 2 before Na (1); $2\,Na + Cl_2 \longrightarrow 2\,NaCl$ (1).	1+1+1
28	Lemon juice: blue litmus turns red (1); Soap solution: red litmus turns blue (1); Phenolphthalein: colourless in lemon juice, pink in soap (1).	1+1+1
29	Lustre (1); Malleable and ductile (1); Good conductors of heat and electricity (1).	1+1+1
30	Methane CH_4 (1); Ethane C_2H_6 (1); Propane C_3H_8 (1).	1+1+1
31	Carbon dioxide from air (1); Water from soil (1); Light from Sun, chlorophyll in chloroplast (1).	1+1+1
	OR: Voluntary: brain-controlled (1); Reflex: spinal cord (1); Reflex is faster (1).	1+1+1
32	Binary fission: Amoeba (1); Budding: Hydra (1); Spore formation: Rhizopus (1).	1+1+1
	OR: Seed shape (round or wrinkled) (1); Seed colour (yellow or green) (1); Plant height (tall or dwarf) (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Combination reaction (1); Product is magnesium oxide (1); To remove the oxide layer (1).	1+1+1+1+1
34	Acid sour, base bitter (1); Acid turns blue litmus red, base turns red litmus blue (1); Acid gives H^+, base gives OH^- (1); Acid pH below 7, base pH above 7 (1); Examples: HCl, NaOH (1).	1+1+1+1+1
35	Lustre: metals lustrous, non-metals dull (1); Conduction: metals conduct (1); Malleability: metals malleable, non-metals brittle (1); Hardness: metals generally hard (1); Oxides: basic vs acidic (1).	1+1+1+1+1
36	CH_4: 4 C-H single bonds (1); C_2H_4: 4 C-H and a C=C (1); H_2O: 2 O-H (1); Shared pairs: 4, 6 and 2 (1); Covalent bonds form by sharing (1).	1+1+1+1+1
	OR: Takes place in chloroplast (1); Chlorophyll absorbs light (1); Water splits (1); CO_2 is reduced to glucose (1); $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (1).	1+1+1+1+1
37	Dendrite receives information (1); Cell body has nucleus (1); Axon carries impulse (1); Nerve endings pass impulse (1); Synapse between neurons (1).	1+1+1+1+1
	OR: Amoeba is a single cell with a nucleus (1); The nucleus divides into two (1); Cytoplasm constricts in the middle (1); Two daughter cells separate (1); Each grows into a new Amoeba (1).	1+1+1+1+1
38	Parents TT x tt (1); F1 is Tt, all tall (1); Gametes of F1 are T and t (1); F2 has TT, Tt, Tt, tt (1); Ratio is 3 tall : 1 dwarf (1).	1+1+1+1+1
	OR: At infinity: image at F, real, point (1); At C: image at C (1); At C: real, inverted, same size (1); Between P and F: behind the mirror (1); Virtual, erect, enlarged (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 46	23 pairs.
2	**(c)** refraction	Definition.
3	**(a)** myopic	Near-sighted.
4	**(a)** its length	R is proportional to length.
5	**(c)** a magnetic field	Oersted's discovery.
6	**(c)** food chain	Linear.
7	**(d)** blue	Hydrated copper(II) ions are blue.
8	**(c)** salt and water	Neutralisation.
9	**(c)** haematite	Haematite is Fe_2O_3.
10	**(c)** allotropes of carbon	Both are pure carbon.
11	**(c)** four	Two atria and two ventricles.
12	**(a)** phototropism	Phototropism.
13	**(c)** sperm	Sperm cells are male gametes.
14	**(b)** Colour of the eyes	Controlled by genes.
15	**(b)** two spherical surfaces	Basic definition.
16	**(a)** two triangular bases and three rectangular lateral surfaces	Glass prism.
17	**(c)** 10 ohm	Add.
18	**(c)** Fleming's left-hand rule	Force direction.
19	**(b)** ultraviolet rays	UV.
20	**(c)** reactants	Reactants are written on the left.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Production of new individuals of the same kind (2).	2
22	Gregor Mendel (2).	2
23	Light bounces back from a polished surface (1); Surface of the mirror (1).	1+1
24	Retina (2).	2
25	Rate of flow of electric charge (2).	2
26	The region around a magnet where its force is felt (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Shaving: concave (1); Torch: concave (1); Rear-view: convex (1).	1+1+1
28	Iris and pupil control light (1); Cornea and lens focus the light (1); Retina receives the image (1).	1+1+1
29	Q = I t (1); Q = 2 x 5 (1); Q = 10 C (1).	1+1+1
30	Closed curves (1); Outside: N to S (1); Inside: S to N (1).	1+1+1
31	Natural: forest, pond, lake (1); Artificial: aquarium, garden, crop field (1); Natural ecosystems are not made by humans (1).	1+1+1
	OR: Combination (1); Two reactants form a single product (1); Exothermic (heat is released) (1).	1+1+1
32	Sour taste (1); Turn blue litmus red (1); React with metals to give hydrogen (1).	1+1+1
	OR: $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (2); Sodium hydroxide and hydrogen (1).	2+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Cornea (1); Iris and pupil (1); Lens and ciliary muscles (1); Retina (1); Optic nerve (1).	1+1+1+1+1
34	Current is proportional to the potential difference at constant temperature (1); V = I R (1); Graph is a straight line through the origin (1); Slope = V / I (1); Slope gives the resistance (1).	1+1+1+1+1
35	They emerge from the north pole and end at the south pole (1); They form closed curves (1); The tangent gives the direction of the field (1); Closeness shows strength (1); No two lines intersect (1).	1+1+1+1+1
36	Ecosystem: living and non-living parts (1); Abiotic: air, water, sunlight (1); Biotic: producers (1); Biotic: consumers (1); Biotic: decomposers (1).	1+1+1+1+1
	OR: Slow attack on a metal by moisture and air (1); Needs air (oxygen) and moisture (1); Rust is $Fe_2O_3 \cdot x\,H_2O$ (1); Prevention: painting, oiling (1); Galvanisation, alloying (1).	1+1+1+1+1
37	$Na_2CO_3 \cdot 10\,H_2O$; prepared from baking soda (1); Uses: glass, soap, hard water (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O$; heat gypsum at 373 K (1); $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Uses: bone plaster, toys (1).	1+1+1+1+1
	OR: Brass: Cu + Zn (1); Bronze: Cu + Sn (1); Solder: Pb + Sn (1); Stainless steel: Fe, Ni, Cr (1); Amalgam: Hg alloy (1).	1+1+1+1+1
38	$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Esterification needs conc. H_2SO_4 (1); Ester smells fruity (1).	1+1+1+1+1
	OR: Xylem: water; phloem: food (1); Xylem: dead cells; phloem living (1); Xylem upward; phloem both ways (1); Xylem passive; phloem active (1); Vessels/tracheids vs sieve tubes (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** soapy	Bases are soapy to touch.
2	**(a)** ductility	Ductility.
3	**(c)** 4	Tetravalent.
4	**(d)** CO_2 and water	CO_2 + H_2O.
5	**(a)** axon	Axon.
6	**(b)** binary fission	Unicellular organisms divide.
7	**(d)** garden pea	Pisum sativum.
8	**(b)** reflection	Definition.
9	**(b)** cornea	Cornea.
10	**(d)** Rubber	No free electrons.
11	**(a)** tesla	T.
12	**(d)** Sunlight	Abiotic.
13	**(a)** decomposition reaction	One reactant gives two products.
14	**(c)** acidic	pH below 7 is acidic.
15	**(b)** potassium	K is at the top.
16	**(b)** ethene	Alkene.
17	**(c)** liver	Liver.
18	**(b)** thyroid	Thyroid.
19	**(c)** stigma	It receives pollen.
20	**(d)** XY	One X and one Y.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Living and non-living components interacting with each other (2).	2
22	A single compound splits into two or more simpler substances (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1).	1+1
23	Colourless in acid (1); Pink in base (1).	1+1
24	Brass: Cu and Zn (1); Bronze: Cu and Sn (1).	1+1
25	Same molecular formula (1); Different structures (1).	1+1
26	Rhythmic contraction of muscles of the alimentary canal (1); Pushes food forward (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Low melting and boiling points (1); Poor conductors of electricity (1); Mostly burn in air (1).	1+1+1
28	Absorbs water (1); Stores undigested food (1); Egestion (1).	1+1+1
29	Insulin: pancreas (1); Thyroxine: thyroid (1); Adrenaline: adrenal (1).	1+1+1
30	Ovary (1); Oviduct (1); Uterus (1).	1+1+1
31	TT gives only T (1); Tt gives T and t (1); tt gives only t (1).	1+1+1
	OR: n = c/v (1); n = 3×10^8 / 1.25×10^8 (1); n = 2.4 (1).	1+1+1
32	Weakening of ciliary muscles (1); Reduced flexibility of the lens (1); Near point recedes with age (1).	1+1+1
	OR: Length (1); Area of cross-section (1); Nature of the material (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Diabetes: insulin (1); Goitre: iodine/thyroxine (1); Dwarfism: growth hormone (1); Gigantism: excess (1); Treatments (1).	1+1+1+1+1
34	Pollen lands on the stigma (1); It grows a pollen tube through the style (1); The tube reaches the ovule in the ovary (1); Male gamete fuses with the female gamete (1); The zygote is formed (1).	1+1+1+1+1
35	Allele: alternative form of a gene (1); Homozygous: same alleles (1); Heterozygous: different alleles (1); F1: first filial generation (1); F2: second filial generation (1).	1+1+1+1+1
36	1/v = -1/20 + 1/40 (1); 1/v = -1/40 (1); v = -40 cm (1); Image at C (1); Real, inverted, same size (1).	1+1+1+1+1
	OR: Sunlight falls on raindrops (1); Refraction at the first surface (1); Dispersion into colours (1); Internal reflection (1); Second refraction and colours reach the eye (1).	1+1+1+1+1
37	E = P t = 0.1 kW x 10 h (1); E = 1 kWh (1); I = P / V (1); I = 100 / 220 (1); I = 0.45 A (1).	1+1+1+1+1
	OR: Principle: electromagnetic induction (1); Armature coil, magnets, slip rings, brushes (1); Coil rotates in the field (1); Current is induced and changes direction every half turn (1); Output is alternating current (1).	1+1+1+1+1
38	Harmful pesticide enters the water (1); Taken by plants (1); Eaten by herbivores (1); Eaten by carnivores (1); Highest concentration at the top level (1).	1+1+1+1+1
	OR: Brown fumes are seen (1); $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ (1); Thermal decomposition (1); Brown gas is nitrogen dioxide (1); Residue is yellow lead oxide (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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